

Question Number	Answer	Additional Guidance			Mark
3(a)	x 2.5				Cleri (1)
Question Number	Answer	Additional Guidance			Mark
3(b)(i) Clip with 3bii	- calculation of observed values (1) - calculation of $\frac{(O - E)^2}{E}$ (1) - calculation of chi-squared value (1)		Observed value	Expected value	Exp (3)
		Smooth light	12	20	
		Rough light	15	20	
		Smooth dark	24	20	
		Rough dark	29	20	
		Accept MP1 one correct substitution into part of calculation Accept MP2 for one value eg 3.20 Accept: 3.20 / 1.25 / 0.80 / 4.05 for MP1 and 2 9.3 / 9.30 / 9.300 correct answer only gains 3 marks Accept ECF for MP2 and 3			

Question Number	Answer	Additional Guidance	Mark
3(b)(ii) Clip with 3bi	An answer including three of the following • calculated value (9.3) is more than a critical value stated/indicated from the table (1) • therefore reject the null hypothesis (1) • top shells show a preference (1) • Suitable comment on preference (1)	Accept any stated value against any critical value with correct reasoning for all MP's e.g. top shells prefer dark to light / no preference for rough or smooth	Exp (3)

Question Number	Answer	Additional Guidance	Mark
3(b)(iii)	An answer which includes three of the following: • leave them for longer so that they have time to {move around / select their preferred habitat} (1) • put more than five molluscs into each area so that there is more data to analyse (1) • controlled temperature so they move {in the same way/at the same speed} (1) • cover the light side with a transparent covering eg glass so that {both sides are covered / you are not comparing covered and uncovered} (1) • use a round tray so that the molluscs do not collect in corners (1)		Exp (3)

(Total for question 3 = 9 marks)

